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Approaches and outcomes of community health worker's interventions for hypertension management and control in low- and middle-income countries: Systematic review

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Approaches and outcomes of community health worker's interventions for hypertension management and control in low- and middle-income countries: Systematic review

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Abstract

Objectives: To critically appraise the scope, content and outcomes of community health worker (CHW) interventions designed to reduce blood pressure(BP) in low and middle income countries (LMICs).

Method: We performed a database search (PUBMED, EMBASE, CINAHL, PsycINFO, OpenGrey, Cochrane Central Trials Register and Cochrane Database of Systematic Reviews) to identify studies in LMICs from 2000 to September 2020. Eligible studies were randomized control trials or quasi experimental studies published in English and reporting CHW interventions geared to the management and prevention of hypertension in LMICs. Two independent reviewers screened the titles, abstracts and full texts of publications for eligibility and inclusion. Relevant information was extracted from these studies using a tailored template.

Results: Of the 960 articles identified, 14 met the predefined criteria. Twelve of the included studies were cluster randomized trials whereas remaining two were interventional pre- and post-studies. The CHW interventions were mainly community based, designed to promote BP control mainly for hypertensive patients but also targeted prevention of high BP in healthy individuals. CHW interventions had positive effects in the BP reduction, linkage to care, treatment adherence, and in reducing CVD risk level among both hypertensive and normotensive individuals.

Conclusions: This review found that linking CHWs with existing public health care infrastructure enhances control and management of BP than the usual care. Programs involving CHWs as multidisciplinary team members hold promise in the management and control of hypertension in LMICs.

Strengths and limitations of this study

- This is the first systematic review focused on approaches and outcomes of CHW interventions in the management and prevention of hypertension in LMICs.
- We conducted a comprehensive searches of databases to ensure that all relevant publications were identified.
- Potential bias in the conduction of this review was minimized by having the authors independently screen through the search results and extract the data autonomously.
- Due to diversity of the study participants, interventions, and outcomes of the studies

INTRODUCTION

Globally, hypertension is a leading modifiable risk factor for cardiovascular disease and premature death[1, 2], with an estimated 1.13 billion people worldwide living with it. Two-thirds of this burdened population live in low- and middle-income countries[3]. The World Health Organization (WHO) targets to have a worldwide 25% reduction in the prevalence of hypertension by the year 2025[4]. Due to an ageing population and increase in lifestyle risk factors such as lack of physical activity, unhealthy diet, smoking and alcohol consumption, the global prevalence of hypertension is increasing[2]. However, there are disparities in these changes of prevalence of hypertension worldwide. While high-income countries (HICs) experienced a modest decrease (2.6%) in hypertension prevalence in the last two decades, the low and middle-income countries (LMICs) experienced significant increase of 7.7% [5]. The proportion of controlled hypertension is also low, especially in LMICs with only 7.7% of patients with hypertension having controlled blood pressure (BP) (<140/90 mmHg) in LMICs[5]. Concerted efforts are urgently needed to combat the emerging hypertension burden in low- and middle-income countries.

Health workers are critical to addressing this emerging burden. In light of critical shortages in the health workforce in LMICs, community health workers (CHW) are increasingly recognized as an essential part of the health workforce needed to achieve public health goals [6-9]. However, the concept of CHWs has no universal definition and has evolved to suit specific contexts, norms, and cultures[10]. In their systematic review '*Who is a Community Health Worker?*' Olaniran et al, 2017 concluded that a single definition may not project the diversity of this group of health workers and emphasized that these are "individuals with an in-depth understanding of the community culture and language, have received standardized job-related training which is of shorter duration than health professionals, and have a primary goal of providing culturally appropriate health

services to the community"[10]. In this review a CHW will refer to "lay health worker carrying out functions related to health care delivery; trained in some way in the context of the intervention and having no formal professional or paraprofessional certificate or degree in tertiary education"[11]. Use of CHWs has been identified as an important strategy in the delivery of culturally relevant programs for hypertension control in different settings. The CHWs approach is not only affordable and sustainable but also enhances cultural relevance of health information, providing an important linkage between community and health care system[12].

Previously, systematic reviews have assessed the effectiveness of CHW community-based programs in the management and prevention of non-communicable diseases[7, 13-15] but few have focused on their in the care of people with hypertension specifically[16, 17]. Evidence from systematic reviews of community based randomized controlled trials and cluster randomized trials from LMICs show that CHWs are effective in altering the risk factors for non-communicable diseases (NCDs) and demonstrated their effectiveness in modification of physical parameters such as BP and cholesterol levels[7]. Although CHW delivered community-based BP screening and education initiatives are recommended for management and prevention of hypertension in LMIC [18], we did not find a systematic review looking at the approaches and outcomes of these interventions. The purpose of this systematic review therefore was to critically appraise the scope, content and outcomes of CHW interventions designed to reduce BP in LMICs.

Review question/objective: The objective of this systematic review was to identify studies reporting on CHW interventions for hypertension management and control and determine the approaches and outcomes of such interventions in LMICs countries. Specifically, we asked: 1) what are the types of CHW interventions in the management and control of BP in LMICs and 2) what are the impacts of CHW interventions on the management and control of BP in LMICs?

METHODS

Protocol and registration: Methods of the review and inclusion were documented in advance, but not published in a review protocol.

Search strategy: We carried out a systematic review of literature and a descriptive review of studies reporting CHW interventions for hypertension management and control. The review followed the recommended methodological framework of conducting a systematic review of healthcare interventions[19] and the reporting adhered to the PRISMA guidelines for reporting such reviews[20]. The relevant studies were identified using different search strategies. We identified peer-reviewed literatures – randomized controlled trials and quasi-experimental studies published between January 2000 and September 2020. We included studies that were available in English language and focused on the management and control or reduction of high BP among adult population in LMICs. We employed Covidence, an online software (Covidence, Roende, Denmark) to extract and screen for articles to be included in the study.

We searched for relevant studies in electronic databases: PUBMED, EMBASE, CINAHL, PsycINFO, OpenGrey, Cochrane Central Trials Register and Cochrane Database of Systematic Reviews. This comprehensive search was needful to identify randomized controlled trials and quasi-experiment studies in LMICs. The search string used towards this review was "Community

Health Workers"[Mesh] OR "community health volunteers" OR "lay health workers") AND "Hypertension"[Mesh]) OR "Cardiovascular Diseases"[Mesh]) OR "Blood Pressure"[Mesh]) AND "Developing Countries"[Mesh] OR "low and middle income countries") AND ("Randomized Controlled Trials as Topic"[Mesh] OR "Non-Randomized Controlled Trials as Topic"[Mesh]). The full search strategies for each of the databases included are provided in the supplementary document (Appendix 1).

Inclusion criteria: Studies were included in the analysis if they met the following;

Population: Studies focusing on management or control of BP among the general population or among pre-hypertensive or hypertensive patients.

Intervention: CHW interventions focused on management and control of hypertension. In the current study, CHW referred to a lay health worker who is not formally and professionally trained, either paid or working on voluntary basis, and carrying out functions associated with health care delivery in the management and control of hypertension.

Comparison: Studies had a comparison control arm of either no intervention, usual care or another intervention

Outcomes: Reduction in BP, proportion of patients with controlled blood pressure, engagement in hypertension care, adherence to hypertensive drugs reduction in cardiovascular risk.

Study designs: Randomized controlled trials (RCTs) and quasi experimental studies were included.

Setting: Only studies conducted in LMICs (as defined by the World bank) were included.

Exclusion criteria: Studies whose intervention was not adequately described or whose outcomes were not clearly described were excluded from this analysis.

Study selection

The inclusion criteria described above informed the study selection. Two reviewers (GWM and KM) assessed all retrieved lists of citations and abstracts independently. The full texts of all potentially eligible studies were retrieved and critically examined to ensure they met all the inclusion criteria. Discrepancies between reviewers about the eligibility of retrieved studies were resolved by discussion. The reference lists of the selected publications were also searched for identification of additional studies.

Extraction of data

Data extraction was done using a tailored data extraction template that was created using guidelines outlined in the Cochrane Handbook for Systematic Reviews of Interventions [21]. Relevant data including the author, year of publication, type of publication and journal published in, study location, study design, sample size and baseline characteristics, intervention features, outcome measurements and key findings were extracted into an Excel sheet template.

Quality assessment

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The quality of the studies that met all the inclusion criteria was assessed using the Cochrane criteria. Specifically, the risk of bias in generation of the randomization sequence, allocation concealment and blinding (participants, personnel and outcome assessors), incomplete outcome data and selective reporting were assessed as high, low or unclear. Two authors (GWM and KM) assessed the risk of bias and disagreements were resolved by consensus.

Ethics and dissemination

This paper involved a synthesis of publicly available articles and therefore did not require ethical approval.

Patient and Public Involvement

We did not involve patients or members of the public in the design, or conduct, or reporting, or dissemination plans of this research.

RESULTS

The search strategy identified 960 records through database searching (n = 934) and reference lists (n = 26). After excluding duplicates, a total of 858 records were reviewed (Figure 1). Out of these, 826 were excluded because information provided in the title and abstract did not meet the inclusion criteria leaving 32 potentially eligible studies for full text review. Of these, 18 did not meet all the inclusion criteria and were excluded for reasons (some more than one), that included not being interventional studies (n=8), not based in low- and middle-income countries (n=5), being protocols of studies without results presented (n=5) and not reporting on BP as an outcome (n=3). Thus, a final 14 studies were used for final data extraction, twelve being cluster randomized trials[22-33] and the other two being interventional pre- and post-studies[34, 35]. The summary characteristics of the included papers is included in Table 1. The study populations were hypertensive patients in 6 studies[25, 26, 28-30, 33], had at least 1 CVD risk factor in 2 studies[24, 32] and in 6 studies the participants involved the general population[22, 23, 27, 31, 34, 35]. All the studies involved adult populations except one study that involved children and young people aged 3-39 years[27]. The duration of the interventions ranged from 3 months to 2 years. Geographically, 9 studies were from South Asia[24, 26, 27, 29-34], 3 from Sub Saharan Africa[22, 23, 28] and 1 from East Asia[35] and 1 from South America[25].

Risk of bias for the included studies

The risk of bias of the included studies is shown in Figure 2. The risk of selection bias as result of sequence generation and allocation concealment was low in 12 (86%) of the studies and high in 2(14%). The risk of performance bias and reporting bias due to lack of blinding of participants, personnel and outcome assessors was low in 8(57%) and high in one, but unclear in 5(36%) of the studies. The risk of attrition bias (loss to follow-up) was low in 11(79%), high in one and unclear in two studies while risk of bias due to selective outcome reporting was low in 11(79%) of the studies and unclear in 3(21%).

Table 1. Summary of papers included

Author, year, journal & setting	Study design	Baseline sample characteristics	Intervention design	Outcome measurement	Key findings
Joshi et al, 2019 American Heart Journal (v.216) Rural India	A cluster randomized trial	**controls** 1611 (567 hypertensive), mean age 61.7±10.4 years, mean SBP 130.3±20.8, mean DBP 79.4±11.7, interheart risk score 10; **Intervention** 1650 (597 hypertensive), mean age 61.7±10.2yrs, mean SBP 130.3±20.5, DBP 78.9±11.3, interheart risk score (10)	CHWs home visitation to encourage prescriptions' adherence and measure BP. CHWs provided short goal-directed printed slogans to promote integration of preventive therapies with activities of daily living.	Primary outcome- Mean SBP. Secondary outcome- inter-heart risk score, and adherence to prescription drugs intake	SBP drop was not impacted Inter-heart risk score dropped to 8.2 (intervention) versus 8.3 (control) CHWs promoted adherence to therapy
Jafar et al. 2009 Annals of Internal Medicine (v.151) Karachi, Pakistan	A cluster randomized trial	1341 hypertensive (SBP ≥140 mmHg or DBP ≥90 mmHg or on antihypertensive and all aged ≥40 years)	Home health education (HHE) by CHWs: 3-monthly HHE, and annual training of general practitioners (GP) in hypertension management.	Primary outcome - change in SBP from baseline to the last follow-up visit (2 years). Secondary outcome - % with <140 mmHg SBP & <90mmHg DBP at last follow up (2 years)	SBP decrease in HHE and GP group (10.8 mm Hg [95% CI, 8.9 to 12.8 mm Hg]) than in the GP-only, HHE-only, or no intervention groups (P<0.001).
Jafar et al, 2020 The New England Journal of Medicine Multicountry: Rural Bangladesh, Pakistan and Sri Lanka	A cluster-randomize controlled trial	2645 hypertensive adults- 1330 intervention group, 1315 control Mean age 58.8±11.5 years 64.3% were female (877 in intervention and 824 in control), 35.7% male (distribution not described), BP was uncontrolled in 69.6% and very poorly controlled in 29.6% of participants SBP (intervention group 146.7±22.4, control group 144.7±21.0 mmHg); DBP (89.1±14.7intervention group and 87.8±13.8 in control)	Home-planned CHWs-led health education (intervention) together with 3-monthly BP monitoring for 24 Months,	Primary outcome - Reduction in SBP at 24 months. Secondary outcomes – Reduction in DBP at 24 weeks. and, % of participants with BP control (SBP<140 mmHg and DBP <90 mmHg)	Mean SBP reduction greater with intervention (by 5.2 mmHg) compared to controls DBP reduction of 2.8 mmHg greater in intervention than in control group. BP control (<140/90 mmHg) achieved in 53.2% of intervention group compared with 43.7% in control group Summary: Linking government trained CHWs with existing public health care infrastructure enhanced BP reduction than in usual care.
Vendatha n et al, 2019 Journal of American College of Cardiology (vol 74) Turbo, Western Kenya	A cluster randomized trial	1460 Participants (58% female, 42% male). **Control** (usual care) 491, **Intervention** paper-based (500), and smartphone (469). Pooled baseline SBP 159.4 mmHg, DBP 89.7±12 mmHg. Usual care (158.4±19.6 & 89.0±12.3), Paper based (158.3±20.3 & 90.4±12.1) and Smart Phone (161.3±18.5 & 89.7±11.5) for SBP & DBP respectively.	Tailored behavioral communication by CHWs using "paper-based" and "smartphone based" approaches.	Primary outcomes - 1) linkage to care, and 2) change in SBP. Secondary outcome – DBP change.	Overall improvement (49%) linkage to care. Significantly greater linkage in the usual care and smartphone arms. Average overall follow-up SBP and DBP were 149.9 and 90.8 mmHg, respectively. Modest improvement in SBP in smart phone than usual care groups (−13.1 mmHg vs. −9.7), **but** difference insignificant. BP control (<140/90) achieved in 26% of participants – Group differences insignificant.
Peiris et al, 2019 PLoS ONE	A stepped-wedge, cluster randomised	**Intervention group** (n=4348, 55.6 % Female). 60.3±10.7 years, 46.8% no formal schooling, 45.7% (1986) hypertensive, Angina 11.8%,	Three, 6-months intervention thus: (1) mobile tablet based CVD risk assessment	Primary outcome – proportion whose SBP drops <140 mmHg.	Increased treatment rates among high risk individuals assessed by CHWs, but, the effects on BP were not demonstrated.

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West Godavari District, Andhra Pradesh, Rural India	controlled trial	stroke 9.2%; 10 year CVD risk SBP > 160 mmHg or DBP > 100mmHg 53.4% (2324); on antihypertensives 41.8%; SBP 156.6±22.9; DBP 89.1±13.8 **Control group** (n=4294) 55.9 % Female). 61.0±10.9 years, 46.9% no formal schooling, 46.5% hypertensive (i.e. 1997), Angina 11.8%, stroke 8.1%; 10 year CVD risk SBP >160 mmHg or DBP >100mmHg 52.4% (2251); on BP drugs 42.3%; SBP 157.3±23.2; DBP 88.8±13.9	by CHWs; (2) electronic referral to PHC doctors; and (3) a tracking system for follow-up care.	Secondary outcomes – difference in mean BP levels, difference in proportion using BP drugs.	
Jafar et al, 2010 British Medical Journal Karachi, Pakistan	A cluster randomised controlled trial	4023 participants, aged 3-39 years – mean age for HHE, the intervention n=2008 was 18.9±9.8, for control n=2015 was 19.0±9.8; males in HHE 46.3% (i.e 929); Males in control 45.8% (i.e 923); mean SBP in HHE 114±14 mmHg, in control 115±14; mean DBP in HHE 74±11, in control also 74±11. HHE literacy 11.5%, in non HHE 13.8%	3-monthly home based family health education by CHWs	Primary outcome – change in SBP after 2 years follow-up. Secondary outcome – Change in DBP between baseline and at 2 years	Family based HHE by CHWs reduced the usual rise in BP with age in children and young adults.
He et al, 2017 The Journal of the American Medical Association Argentina (area not specified)	A cluster randomized trial	1432 hypertensive adults Mean age intervention group (n=743) age 56.1±13.6 years, control (n=689) 55.5±13.0 years. 52.6% and 53.4% of intervention and control groups respectively were female, Mean SBP 151.7±16.8 and 149.8±15.5 mmHg and DBP, 92.2±12.2 and 90.1±12.9 among intervention and control groups respectively. 86% in intervention group and 84% in control group were on BP drugs	multicomponent intervention: (1) CHW–led home health coaching, BP monitoring, BP audit and feedback, and (2) a GP intervention, and a (3) text messaging intervention over 18 months. Controls receiving usual care.	Primary outcomes of differences in SBP and DBP between baseline and follow-up end Secondary outcomes were % with controlled BP <140/90mmHg	CHWs helped low-income patients with uncontrolled hypertension achieve greater decrease in SBP and DBP through multicomponent interventions compared to patients receiving usual care over 18 months.
Khetan etal 2019 Global Heart journal Dalkhola, India	A Cluster Randomized Controlled Trial	1242 adults (35 to 70 years drawn from 12 clusters with at least 1 CVD risk factor (hypertension 650, diabetes 317, smoking 500) Intervention group- 736 Mean age 52.1±9.6 years Control group- 506 Mean age- 51.7 ± 9.8 years.	Behavioral change communication and BP monitoring through regular CHWs home visits once every 2 months for 2 years. The control group received usual care in the community	Primary outcomes were change in SBP, FBG and self-reported number of daily cigarettes from visit 1 to post-intervention among hypertensives, diabetics and smokers.	Improved SBP in hypertension, and inconclusive effect on FBG in diabetes, and no demonstrable effect on smoking.
Setiawan, D & Sargowo ,D 2018 European Society of cardiology congress Batu City, Indonesia	Interventional pre and post-test study	102 subjects (Men 21; Women 81) aged 30 to 75. Mean Framingham Risk Score (FRS) of 6.17 65.75	Community based primary intervention that included 60-minutes of heart exercise thrice weekly for 3-months, and regular health counselling and education on lifestyle changes	Primary outcome – change in mean value of FRS	A community-based health education and physical activity program significantly improved the FRS of a farmer's population. Mean FRS changed from 6.17 65.75 to 4.16 63.92 post-intervention group (Z score = -6.009; p =0.00).
Sankaran et al 2016 Journal of general	Interventional pre and post study	598 patients over the age 60 were screened. Baseline hypertension prevalence rate of 44%.	CHW trained for 7 months offered a community intervention for 1 year, that involved diagnosis,	Primary outcome- BP control at 1 year	After 1 year, 51.8% of hypertensive patients had BP <140/90 mmHg. A teaching program educating community health workers about hypertension followed by

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internal medicine Rural region of Southern India			management and referral of patients with uncontrolled BP and provision of continuing medications and lifestyle advice to those with well-controlled hypertension. GPs did initial evaluation and medication while nurses supervised CHWs, weekly.		screening, diagnosis, and management of hypertension by CHW in the field is a feasible mechanism for achieving hypertension control in a remote tribal community in India.
Gamage et al 2020 Plos One-Medicine South India	A cluster randomised controlled trial	2,382 hypertensive patients. Intervention group- 637 from 5 clusters Mean age-56.6±14.3 Mean SBP 140.5±22.7 Mean DBP 80.4±13.7 Controlled hypertension- 43.5 On hypertensive drugs-38% Control group-1097 form 10 clusters Mean age-56.9 Mean SBP 137.8 ±22.2 Mean DBP 80.6 ±13.9 Controlled hypertension-50.1% On hypertensive drugs -40.6%	Trained CHWs delivered group-based intervention to hypertensive people, consisting of 6 fortnightly sessions of ~90 minutes held within participants' villages (clusters). Sessions included BP monitoring, education about hypertension, and support for healthy lifestyle change.	Primary outcome-change from baseline in the proportion of people with controlled hypertension (BP < 140/90 mm Hg) Secondary outcome-change in SBP and DBP from baseline	BP declined an average of 5.0/2.1 mm Hg more in the intervention group than the usual care group, and control of BP improved from baseline to follow-up, more in the intervention (from 227 [49.5%] to 320 [69.7%] individuals) than in the control group (from 528 [52.2%] to 624 [61.7%] individuals) (odds ratio [OR] 1.6, 95% CI 1.2–2.1; P = 0.001)
Neupane et al 2018 Lancet Glob Health Nepal	An open-label, cluster-randomised trial	1468 with mean BP of 122/80 mm Hg. 16% of participants were smokers, 11% were drinking alcohol in amounts harmful to their health, and 5% had low physical activity. Intervention group 425 normotensive 175 pre-hypertensive 255 hypertensive Control 305 normotensive 128 pre-hypertensive 180 hypertensive	A lifestyle intervention led by female community health volunteers (FCHVs) or usual care (control group). 43 FCHVs provided home visits every 4 months for lifestyle counselling and BP monitoring for a period of 1 year.	Primary outcome- mean systolic blood pressure at 1 year.	The intervention was effective for reduction of BP in individuals with hypertension and ameliorates age-related increases in BP in adults without hypertension in the general population. The mean SBP at 1 year was significantly lower in the intervention group than in the control for all cohorts (–2·28 mm Hg (95% CI –3·77 to –0·79, p=0·003) for normotensive participants, –3·08 mm Hg (–5·58 to –0·59, p=0·015) for prehypertensive participants, and –4·90 mm Hg (–7·78 to –2·00, p=0·001) among hypertensives.
Cappuccio et al 2006 BMC Public Health Ashanti region of central Ghana	A community-based cluster randomised trial	1,013 participants from 12 villages (628 women, 481 rural dwellers). Intervention group Mean age-54±11 years Mean BMI- 21±4 Mean SBP- 129±25 Mean DBP- 77± 13 Hypertensive 30% Control group Mean age-55±11 years Mean BMI- 21±4 Mean SBP- 127±27 Mean DBP- 76± 13 Hypertensive –28 %	Community-based trial of health promotion for population-wide reduction in salt intake, to lower population BP. Health promotion targeted villagers willing to attend and was provided through daily sessions in week one followed by 6-months' once weekly sessions in communal places (church & schools)	Primary outcome-change in mean BP and urinary sodium.	At six months the intervention group showed a reduction in systolic (2.54 mmHg [-1.45 to 6.54]) and diastolic (3.95 mmHg [0.78 to 7.11], p = 0.015) BP when compared to control. There was no significant change in UNa.

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Goudge et al 2017 BMJ Global health Mpumalanga Province, South Africa	A pragmatic cluster randomised controlled trial with repeated cross-sectional surveys.	Control group - 1908 adults with mean age of 56.4±19.8 years, females -56% No hypertension-53.7% Controlled hypertension on medication- 10.2% Uncontrolled hypertension on medication- 9.2% Not on treatment- 27% Intervention group - 1505 adults Mean age-56.81±9.7 years Females -55% No hypertension-53% Controlled hypertension on medication- 6.6% Uncontrolled hypertension on medication- 8.8% Not on treatment- 31.6%	Intervention consisted of two LHWs assigned to each clinic to support nurses over for 18 months in management of patients with chronic disease through booking appointments, retrieving and filing patient files, health education, taking vital signs, prepacking of medication and texting reminders for appointments	Primary outcome - change of proportion of the population who had uncontrolled hypertension with a risk profile indicating at least moderate risk of CVD. Secondary outcomes - changes in proportion of the population with undiagnosed hypertension, had their BP measured, were using medication for hypertension and retention in care of people with diagnosed hypertension defined.	There was no reduction in those with moderate or greater CVD risk but there was improved adherence. The intervention did not improve BP control, despite its success in increasing the number of patients with hypertension attending the clinic, as well as the number that attended on their appointed day.

CVD-Cardiovascular disease; CHWs- Community health workers; BP- Blood pressure; BMI- Body mass index DBP- Diastolic blood pressure; SBP- Systolic Blood pressure; FBG- Fasting blood glucose; FCHVs- Family community health volunteers; Framingham Risk Score; GP- General practitioner; HHE-Home health education; LHWs- Lay health workers; UNa –Urinary sodium

Recruitment and training of CHW

While all the studies met the inclusion criteria of CHW defined as a lay health worker with basic training to supply community members access to health and social services, the recruitment and training of the same was different across the studies. In all of the studies CHW were persons with at least 8 to 10 years of education drawn from the local community but in some of the studies [31, 34] the CHW were specifically female. Most of the studies [28-30, 32] utilized government CHW already working in the health care system while other studies [23, 24, 26, 27, 31, 33] recruited CHW specifically for the study following same criterion used in recruiting the Government CHWs.

The CHW underwent various forms of training to be able to carry out their roles in the studies. The training period ranged from 2 days - 6 weeks with ongoing support from the health care workers. In some of the studies the trainings were shorter ranging from 2-5 days [25, 29, 31, 32] while others ranged from 4-6 weeks [24, 26, 27, 33, 34]. In some of the studies the training period was staggered throughout the study period [24, 30, 34]. In one of the studies the length of the training period for CHW was not specified [28]

Similarly, the content of the training was different across the studies. The content of training included, home health education on CVD risks and behavior change communication strategies [28, 30, 31, 33], basics of hypertension and assessments of CVD risk factors to include measurements technique for BP monitoring and anthropometric measurements [24-26, 29, 31, 32, 34], as well as referral procedure for hypertensive patients [31] survey methods [26]. The recruitment and the CHW training details was not reported in 2 of the studies [22, 23].

Content and Approaches of the CHWs interventions.

The CHWs intervention were designed to promote BP control mainly for hypertensive patients and those with CVD risks but also targeted prevention of high BP in healthy individuals. The interventions were either home-based or community-based with an exception of one study [23] which incorporated the use of CHWs in the primary health clinic to assist nurses in the management of hypertension.

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The content of the activities implemented by the CHWs during the interventions was similar across many of the studies. In addition to the screening or monitoring of hypertension through BP measurements and audits which was the main outcome measure in this review, the CHW were also involved in other activities geared to the management and control of hypertension. First, the major activity in the CHW interventions as outlined in table 1 was health promotion through home-based health education and lifestyle counselling[24-28, 30, 31, 33] and community level behavior change communication[22, 29, 34, 35]. Health education included teaching about hypertension, motivation to engage in care and support for healthy lifestyle change such as reduction in salt intake. Secondly, in one of the study[32] CHW were involved in CVD risk assessment using a mobile tablet-based CVD risk assessment tool. Thirdly, in addition to the health education and BP monitoring, one trial incorporated physical activity component of 60 minutes of heart exercise three times a week for a period of 3 months[35]. Fourth, the CHWs provided follow-up for hypertensive patients and supported patients by monitoring treatment adherence in one of the trials[26]. Finally, in some studies CHWs were utilized as mediators between patients and the healthcare system by providing referral and encouraging linkage to care for those requiring hypertensive treatment [28, 32, 34].

Outcomes and main findings of the studies

The outcomes for the CHWs interventions were different for various studies as shown in Table 1. The reported primary outcome in most of the studies[22, 24-28, 30, 31] was changes in mean SBP while other studies [23, 29, 32, 34] reported proportion of participants with controlled BP (SBP<140 mmHg and DBP <90 mmHg) as a primary outcome. There was significant reduction in the mean SBP in seven studies[22, 24, 25, 27, 30, 31, 33] while in two of the studies [26, 28] the reduction in mean SBP was insignificant. Two of the RCTs reported a reduction in the usual rise in BP with age among children and young adults[27] and normotensive adults in the general population[31]. Similarly, three studies [29, 30, 34] reported significant improvement in the proportion of participants with controlled BP as result of the CHW interventions while three other studies[23, 28, 32] did not find any significant improvement in the BP control for participants. It is notable that even for studies which reported negative results in the control of BP, there were positive effect on linkage to care[28, 32] , treatment adherence [23, 26] and inter-heart risk score[26]. In other studies there was improvement in the primary outcome of mean change in Framingham risk score(FRS) [35], and reduction in the proportion of patients with moderate or high CVD risk level[23].

Other outcomes reported in the studies were changes in DBP[28-30, 33], changes in urinary sodium[22], proportion of the population with undiagnosed hypertension, proportions who had their BP measured and those retained in care among the diagnosed hypertensive patients [23], as well as difference in the proportion of patients on hypertension treatment [23, 32]. In one RCT that had participants with at least one CVD risk factor, FBS among diabetes patients and mean number of daily cigarettes among smokers were reported as primary outcomes in addition to change in SBP[24]. In this study, CHWs interventions had a positive effect on SBP but had inconclusive effect on FBS and mean number of cigarettes smoked[24].

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DISCUSSION

Weak health care systems and a rising burden of CVD calls for health system innovations in the management of the CVD in LMICs. CHWs have been suggested as a way to achieve that goal, with their potential ability to quickly integrate into the health care system with short training periods, they can provide community-based care that is cost-effective and achieve high-quality outcomes [7]. Our review examined the nature and the role CHWs can play in management of hypertension in LMIC. This review found that, in general, studies that linked CHWs with existing public health care infrastructure enhanced control and management of BP over and above usual care conditions. The CHW interventions had positive effects in the reduction of BP, linkage to care, treatment adherence, and in reducing CVD risk level among both hypertensive patients as well as normotensive patients across various LMICs. This underscores the important role CHWs would play in the management of hypertension yet at cheaper cost in countries with leaner economic muscle[36-38]. The reviewed works further show the diverse roles all linked to hypertension care that the CHWs could be involved in ranging from preventive, therapeutic and health systems utilization.

With the growing burden of NCDs, primary prevention is one of major pillars in the control of these diseases in LMICs[39]. In light of critical shortages in the health workforce in LMICs, CHWs provide cheaper alternative that would form the backbone of most primary health care services geared towards management and control of both communicable and non-communicable disease[40]. The studies in this review demonstrate that task-shifting of roles such as health education, interpersonal communication on lifestyle modifications with a focus on primary prevention, screening for early diagnosis and supporting self-management behavior such as in treatment and appointment adherence to CHWs can lead to positive health outcomes in both hypertensive patients and the normotensives drawn from general population. Studies from developing countries show that CHWs have high acceptability at local communities' level which, may contribute the dual benefit of providing both cheaper and acceptable labour in addition to the direct contribution in BP control [41].

While trained CHW may not perform in the same capacity as trained nurses and health educators, with appropriate and specifically focused training and supervision they can successfully contribute to the care of community members who have hypertension, as demonstrated by many of the studies reviewed. The reviews showed that CHWs intervention entailed screening, promotion and monitoring of BP and its control strategies at both home and community levels where only basic primary health care approaches were required. The basics required here could be easy and faster to teach with no need for unnecessary scientific details and the favorable study results from the current review show that such training as the CHWs received is enough to yield demonstrable behavior change and BP modulation outcomes[22-35].

There is general agreement in the reviewed literature that CHWs can have significant reduction of BP and hypertension as a key modifiable risk factor for cardiovascular disease and premature death [1, 2], especially in the LMICs[3]. This review shows that enjoining CHWs in the screening and care of hypertensive patients from such countries has positive results in care of hypertensive patients. That most of the studies in this review had low risk of selection bias in both sequence generation and allocation concealment through randomization (11 studies), had low performance bias because they had both participants and investigators blinded (8 studies), had blinding of

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assessors and so detection bias was minimal (9 studies), and had selective reporting bias minimized at conceptualization and implementation of the protocol as opposed to allowing a direct influence of the outcomes at dissemination stage (11 studies) strengthens the evidence of our review. As such, the evidence deduced on importance of CHWs in mitigation of hypertension becomes especially critical for countries with less resources at a time they report an ageing population, an increase in sedentary lifestyles and other lifestyle risk factors, and a concomitant direct increase in proportions of the general population who are hypertensive[2]. The literature suggests a way out of the reported higher incidence of hypertension in these countries and the high numbers being identified to have poorly controlled BP. The CHWs drawn from these communities would advance the care needed and supplement the lean professional workforce available. CHWs have cultural understanding crucial in providing the necessary linkage between community and health care system [12], and is demonstrated by the current review that their roles can be diverse yet effective show how crucial their acceptance and adoption as part of the necessary hypertension care team could be.

Study limitations and strengths

Our study represents the first synthesis of approaches and outcomes of CHW interventions in the management and control of hypertension in Low and middle income countries. The strengths of the current study are that we conducted comprehensive searches of databases to ensure that all relevant publications were identified. We also reduced potential bias in the conduction of this review by having the authors independently screen through the search results and extract the data autonomously. The study had some limitations however. Due to diversity of the study participants, interventions, and outcomes of the studies included in the review, a quantitative synthesis (meta-analysis) to the overall effect of CHW intervention was not possible and therefore outcomes were summarized for individual studies.

Conclusions

In conclusion the review suggests that CHWs interventions linked with support and/or supervision from health care workers provides a promising avenue for achieving improvements in hypertension control in low- and middle-income countries. The current swelling evidence points to a need to adopt this and integrate it in the care of patients with hypertensive diseases and probably too those requiring linkage to care, treatment adherence, or for overall reduction in CVD risks regardless of hypertension diagnosis status. Such a roll-out will improve BP values for both hypertensive and normotensive individuals at costs low- and middle-income countries can afford.

Conflicts of Interest

The authors declare that they have no competing interests.

Availability of data and materials

No additional data are available.

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Authors contribution

GWM conceived the study idea. GWM and KM performed the literature search. GWM and KM performed data extraction and evidence synthesis. GWM wrote the first draft of the manuscript. JAP and KM critically revised the manuscript. GWM, KM and JAP approved the final version of the manuscript.

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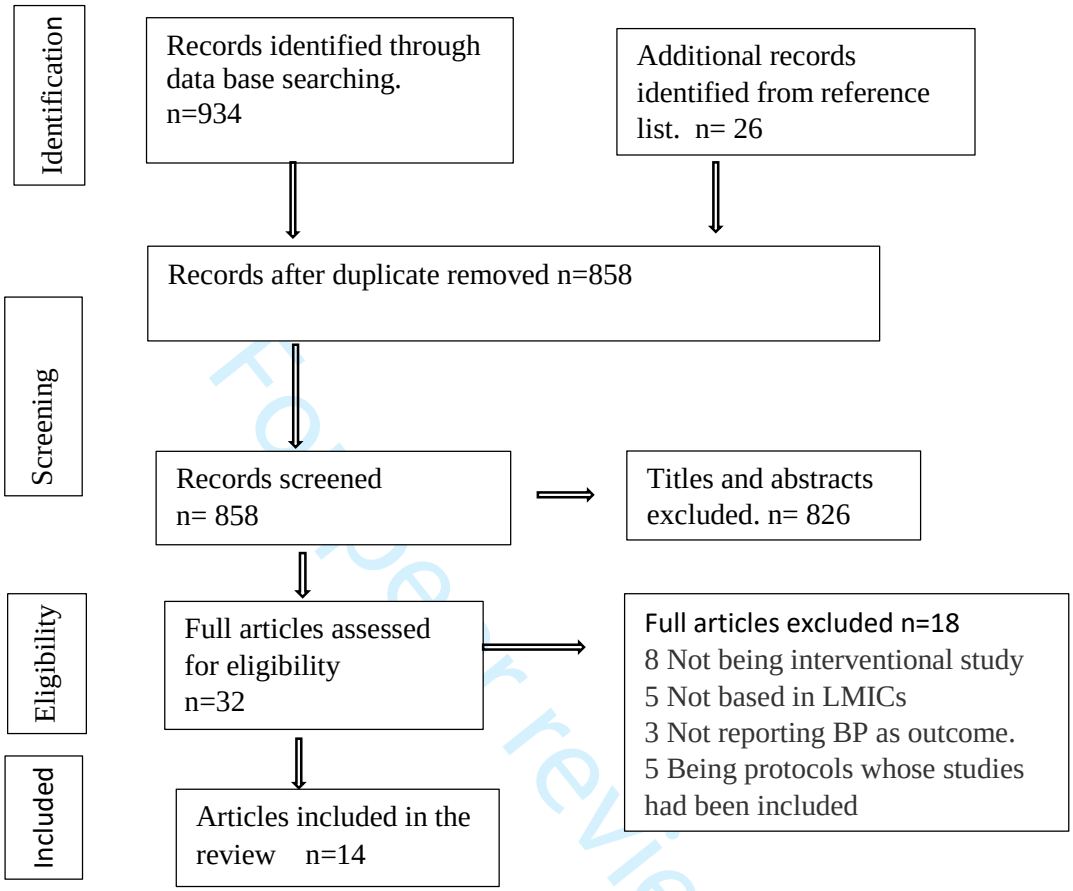


Figure 1. PRISMA Flow diagram showing study selection

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	Sequence generation (selection bias)	Allocation concealment (selection bias)	Blinding of participants and personnel for all outcomes(performance bias)	Blinding outcome assessors(detection bias)	Incomplete outcome data for all outcomes (attrition bias)	Selective outcome reporting(reporting bias)
Cappuccio et al 2006	-	+	+	+	-	+
Gamage et al 2020	+	+	+	-	+	+
Goudge et al 2018	+	+	?	?	+	+
He et al 2017	+	+	+	+	+	+
Joshi et al 2019	+	+	+	+	+	+
Jafar 2009	+	+	+	+	-	?
Jafar 2010	+	?	+	+	-	+
Jafar et al 2020	+	+	+	+	+	+
Khetan et al 2019	+	+	?	+	+	+
Neupane 2018	+	+	-	?	+	?
Peiris et al 2019	+	+	+	+	+	+
Setiawan, D & Sargowo, D 2018	-	-	?	+	?	?
Sankran et al 2016	-	-	?	?	?	+
Vedanthan et al 2019	+	+	?	?	+	+

- High risk of bias
+ Low risk of bias
? Unclear risk of bias

Figure 2: Risk for bias assessment

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Appendix l: Search strings

Database and date	Yield	duplicates	Search string used
PubMed 11/12/20	435	6	"Community Health Workers"[Mesh] OR "community health volunteers" OR "lay health workers") AND "Hypertension"[Mesh]) OR "Cardiovascular Diseases"[Mesh]) OR "Blood Pressure"[Mesh]) AND "Developing Countries"[Mesh] OR "low and middle income countries") AND ("Randomized Controlled Trials as Topic"[Mesh] OR "Non-Randomized Controlled Trials as Topic"[Mesh])
CINAHL 11/12/20	151	3	"Community Health Workers"[Mesh] OR "community health volunteers" OR "lay health workers") AND "Hypertension"[Mesh]) OR "Cardiovascular Diseases"[Mesh]) OR "Blood Pressure"[Mesh]) AND "Developing Countries"[Mesh] OR "low and middle income countries") AND ("Randomized Controlled Trials as Topic"[Mesh] OR "Non-Randomized Controlled Trials as Topic"[Mesh])
PsycINFO 20/12/20	34	0	"Community Health Workers"[Mesh] OR "community health volunteers" OR "lay health workers") AND "Hypertension"[Mesh]) OR "Cardiovascular Diseases"[Mesh]) OR "Blood Pressure"[Mesh]) AND "Developing Countries"[Mesh] OR "low and middle income countries") AND ("Randomized Controlled Trials as Topic"[Mesh] OR "Non-Randomized Controlled Trials as Topic"[Mesh])
Cochrane 20/12/20	198	10	Community health workers [Mesh] and hypertension
EMBASE 20/12/20	116	1	('community'/exp OR community) AND ('health'/exp OR health) AND ('worker'/exp OR worker) AND ('cardiovascular'/exp OR cardiovascular) AND ('diseases'/exp OR diseases) AND ('intervention'/exp OR intervention)
Total	934	20	
Other sources	26		
52 duplicates removed by endnote A further 30 duplicates removed by covidence to give a total of 858			

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Reporting checklist for systematic review and meta-analysis.

Based on the PRISMA guidelines.

Instructions to authors

Complete this checklist by entering the page numbers from your manuscript where readers will find each of the items listed below.

Your article may not currently address all the items on the checklist. Please modify your text to include the missing information. If you are certain that an item does not apply, please write "n/a" and provide a short explanation.

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In your methods section, say that you used the PRISMAreporting guidelines, and cite them as:

Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group. Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement

		Reporting Item	Page Number
Title			
	#1	Identify the report as a systematic review, meta-analysis, or both.	1
Abstract			
Structured summary	#2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number	1-2
Introduction			
Rationale	#3	Describe the rationale for the review in the context of what is already known.	2

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Objectives	#4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	3
Methods			
Protocol and registration	#5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address) and, if available, provide registration information including the registration number.	3
Eligibility criteria	#6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rational	4
Information sources	#7	Describe all information sources in the search (e.g., databases with dates of coverage, contact with study authors to identify additional studies) and date last searched.	3-4
Search	#8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	4
Study selection	#9	State the process for selecting studies (i.e., for screening, for determining eligibility, for inclusion in the systematic review, and, if applicable, for inclusion in the meta-analysis).	4
Data collection process	#10	Describe the method of data extraction from reports (e.g., piloted forms, independently by two reviewers) and any processes for obtaining and confirming data from investigators.	4
Data items	#11	List and define all variables for which data were sought (e.g., PICOS, funding sources), and any assumptions and simplifications made.	4
Risk of bias in individual studies	#12	Describe methods used for assessing risk of bias in individual studies (including specification of whether this was done at the study or outcome level, or both), and how this information is to be used in any data synthesis.	4-5
Summary measures	#13	State the principal summary measures (e.g., risk ratio, difference in means).	n/a

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Planned methods of analyis	#14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I2) for each meta-analysis.	n/a
Risk of bias across studies	#15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	4-5
Additional analyses	#16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	n/a
Results			
Study selection	#17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	5
Study characteristics	#18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citation.	5-9
Risk of bias within studies	#19	Present data on risk of bias of each study and, if available, any outcome-level assessment (see Item 12).	5
Results of individual studies	#20	For all outcomes considered (benefits and harms), present, for each study: (a) simple summary data for each intervention group and (b) effect estimates and confidence intervals, ideally with a forest plot.	6-10
Synthesis of results	#21	Present the main results of the review. If meta-analyses are done, include for each, confidence intervals and measures of consistency.	9-10
Risk of bias across studies	#22	Present results of any assessment of risk of bias across studies (see Item 15).	5
Additional analysis	#23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	n/a
Discussion			
Summary of Evidence	#24	Summarize the main findings, including the strength of evidence for each main outcome; consider their relevance to	11-12

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		key groups (e.g., health care providers, users, and policy makers	
Limitations	#25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review level (e.g., incomplete retrieval of identified research, reporting bias).	12
Conclusions	#26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	12
Funding			
Funding	#27	Describe sources of funding or other support (e.g., supply of data) for the systematic review; role of funders for the systematic review.	13

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Approaches and outcomes of community health worker's interventions for hypertension management and control in low- and middle-income countries: Systematic review

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Approaches and outcomes of community health worker's interventions for hypertension management and control in low- and middle-income countries: Systematic review

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Abstract

Objectives: To critically appraise the scope, content and outcomes of community health worker (CHW) interventions designed to reduce blood pressure (BP) in low- and middle-income countries (LMICs).

Method: We performed a database search (PUBMED, EMBASE, CINAHL, PsycINFO, OpenGrey, Cochrane Central Trials Register and Cochrane Database of Systematic Reviews) to identify studies in LMICs from 2000 to 2020. Eligible studies were interventional studies published in English and reporting CHW interventions for management of BP in LMICs. Two independent reviewers screened the titles, abstracts and full texts of publications for eligibility and inclusion. Relevant information was extracted from these studies using a tailored template. Risk of bias was assessed using the Cochrane collaboration risk of bias tool. Qualitative synthesis of results was done through general summary of the characteristics and findings of each study. We also analyzed the patterns of interventions and their outcomes across the studies. Results were presented in form of narrative and tables.

Results: Of the 1557 articles identified, 14 met the predefined criteria. Of these, 12 were cluster randomized trials whereas two were pretest/posttest studies. The CHW interventions were mainly community-based and focused on behavior change for promoting BP control among hypertensive patients as well as healthy individuals. The interventions had positive effects in the BP reduction, linkage to care, treatment adherence, and in reducing Cardivascular-disease risk level.

Discussion and conclusion: The current review is limited in that, a meta-analysis to show the overall effect of CHW interventions in the management of hypertension was not possible due to the diversity of the interventions, and outcomes of the studies included in the review. Summarized outcomes of individual studies showed CHW enhanced the control and

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management of hypertension. Further studies are needed to indicate the impact and cost-effectiveness of CHW-led interventions in the control and management of hypertension in LMICs.

Funding: This research was supported by the Consortium for Advanced Research Training in Africa (CARTA).

Registration: Methods of the review and inclusion were documented in advance, but not published in a review protocol.

Strengths and limitations of this study

- This is the first systematic review with focus on approaches and outcomes of CHW interventions in the management and prevention of hypertension in LMICs.
- We conducted a comprehensive search of databases to ensure that all the relevant publications were identified.
- Potential bias in the conduction of this review was minimized by having the authors independently screen the search results and extract the data autonomously.
- Due to diversity of the study participants, interventions, and outcomes of the studies included in the review, a quantitative synthesis (meta-analysis) to show the overall effect of CHW interventions was not possible and therefore this review has reported summarized outcomes for individual studies.

INTRODUCTION

Globally, hypertension is a leading modifiable risk factor for cardiovascular disease and premature death[1, 2], with an estimated 1.13 billion people worldwide living with it. Two-thirds of this burdened population live in low- and middle-income countries[3]. The World Health Organization (WHO) targets to have a worldwide 25% reduction in the prevalence of hypertension by the year 2025[4]. Due to an ageing population and increase in lifestyle risk factors such as lack of physical activity, unhealthy diet, smoking and alcohol consumption, the global prevalence of hypertension is increasing[2]. However, there are disparities in these changes of prevalence of hypertension worldwide. While high-income countries (HICs) experienced a modest decrease (2.6%) in hypertension prevalence in the last two decades, the low and middle-income countries (LMICs) experienced significant increase of 7.7% [5]. The proportion of controlled hypertension is also low, especially in LMICs with only 7.7% of patients with hypertension having controlled blood pressure (BP) (<140/90 mmHg) in LMICs[5]. Concerted efforts are urgently needed to combat the emerging hypertension burden in LMICs.

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Health workers are critical to addressing this emerging burden. In light of critical shortages in the health workforce in LMICs, community health workers (CHW) are increasingly recognized as an essential part of the health workforce needed to achieve public health goals [6-10]. However, the concept of CHWs has no universal definition and has evolved to suit specific contexts, norms, and cultures[11]. In their systematic review '*Who is a Community Health Worker?*' Olaniran et al, 2017 concluded that a single definition may not project the diversity of this group of health workers and emphasized that these are "individuals with an in-depth understanding of the community culture and language, have received standardized job-related training which is of shorter duration than health professionals, and have a primary goal of providing culturally appropriate health services to the community"[11]. In this review a CHW will refer to "lay health worker carrying out functions related to health care delivery; trained in some way in the context of the intervention and having no formal professional or paraprofessional certificate or degree in tertiary education"[12]. Use of CHWs has been identified as an important strategy in the delivery of culturally relevant programs for hypertension control in different settings. The CHWs approach is not only affordable and sustainable but also enhances cultural relevance of health information, providing an important linkage between community and health care system[13].

Previously, systematic reviews have assessed the effectiveness of CHW community-based programs in the management and prevention of non-communicable diseases (NCDs)[7, 14-16] but few have focused on the care of people with hypertension specifically[17, 18]. Evidence from systematic reviews of community based randomized controlled trials and cluster randomized trials from LMICs show that CHWs are effective in altering the risk factors for NCDs and demonstrated their effectiveness in modification of physical parameters such as BP and cholesterol levels[7]. Although CHW delivered community-based BP screening and education initiatives are recommended for management and prevention of hypertension in LMIC[19], there are few systematic reviews focused in this area. Reviews have demonstrated the effectiveness of training programs for CHW[20] and task-sharing with non-physician health workers [21] in the management of hypertension in LMICs. Similarly, in their systematic review examining implementation strategies for the control of BP, Mills et al., 2017 found that multilevel and multicomponent strategies such as team-based care involving non-physician health care workers [22] were the most effective in the management of hypertension. The purpose of this systematic review was to critically appraise the scope, content and outcomes of CHW interventions designed to reduce BP in LMICs.

Review question/objective: The objective of this systematic review was to identify studies reporting on CHW interventions for hypertension management and control and determine the approaches and outcomes of such interventions in LMICs. Specifically, we asked: 1) what are the types of CHW interventions in the management and control of BP in LMICs and 2) what are the outcomes of CHW interventions on the management and control of BP in LMICs?

METHODS

We carried out a systematic review of literature and a descriptive review of studies reporting CHW interventions for hypertension management and control. The review followed the recommended

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methodological framework of conducting a systematic review of healthcare interventions[23] and the reporting adhered to the PRISMA 2020 guidelines for reporting such reviews[24].

Eligibility Criteria: We identified peer-reviewed literature – randomized controlled trials, quasi-experimental and pretest/posttest studies published between January 2000 and September 2020. We included studies that were available in English language and focused on the management and control of high BP among adult population in LMICs.

Inclusion criteria: Studies were included in the analysis if they met the following;

Population: Studies focusing on management or control of BP among the general population or among pre-hypertensive or hypertensive patients.

Intervention: CHW interventions focused on management and control of hypertension. In the current study, CHW referred to a lay health worker who is not formally and professionally trained, either paid or working on voluntary basis, and carrying out functions associated with health care delivery in the management and control of hypertension.

Comparison: Studies had a comparison control arm of either no intervention, usual care or another intervention. However, pretest/posttest studies with a well-defined intervention were also included.

Outcomes: Reduction in BP, proportion of patients with controlled BP, engagement in hypertension care, adherence to hypertensive drugs and reduction in cardiovascular risk.

Study designs: Randomized controlled trials (RCTs), quasi experimental and pretest/posttest studies were included.

Setting: Only studies conducted in LMICs (as defined by the World bank) were included.

Exclusion criteria: Studies whose intervention was not adequately described or whose outcomes were not clearly described were excluded from this analysis.

Information sources: We searched for relevant studies in electronic databases: PUBMED, EMBASE, CINAHL, PsycINFO, OpenGrey, Cochrane Central Trials Register and Cochrane Database of Systematic Reviews. This comprehensive search was needful to identify interventional studies meeting the eligibility criteria in LMICs.

Search strategy: The relevant studies were identified using different search strategies. The basic search string used towards this review was "Community Health Workers"[Mesh] OR "community health volunteers" OR "lay health workers") AND "Hypertension"[Mesh]) OR "Cardiovascular Diseases"[Mesh]) OR "Blood Pressure"[Mesh]) AND "Developing Countries"[Mesh] OR "low and middle income countries") AND ("Randomized Controlled Trials as Topic"[Mesh] OR "Non-Randomized Controlled Trials as Topic"[Mesh]). Other alternative terms were added during the search. The full search strategies for each of the databases included are provided in the supplementary document (Appendix 1).

Study selection

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We employed Covidence, an online software (Covidence, Roende, Denmark) to extract and screen for articles to be included in the study. The inclusion criteria described above informed the study selection. Two reviewers (GWM and KM) assessed all retrieved lists of citations and abstracts independently. The full texts of all potentially eligible studies were retrieved and critically examined to ensure they met all the inclusion criteria. Discrepancies between reviewers about the eligibility of retrieved studies were resolved by discussion. The reference lists of the selected publications were also searched for identification of additional studies.

Data extraction and risk of bias assessment

Data extraction was done using a tailored data extraction template that was created using guidelines outlined in the Cochrane Handbook for Systematic Reviews of Interventions [25]. Relevant data including the author, year of publication, type of publication and journal published in, study location, study design, sample size and baseline characteristics, intervention features, outcome measurements and key findings were extracted into an Excel sheet template.

The quality of the studies that met all the inclusion criteria was assessed using the Cochrane collaboration risk of bias tool[26]. Specifically, the risk of bias in generation of the randomization sequence, allocation concealment and blinding (participants, personnel and outcome assessors), incomplete outcome data and selective reporting were assessed as high, low or unclear. Two authors (GWM and KM) assessed the risk of bias and disagreements were resolved by consensus.

Synthesis of the results: Qualitative synthesis of all the studies that met the inclusion criteria described above was done by summarizing the characteristics and findings of the individual studies. The study characteristics and findings were summarized in form of a table. We used text to describe different strategies used in the recruitment, training of CHW and in the implementation of interventions for control of hypertension. We characterized the CHW intervention by examining the recruitment process for CHW- whether drawn from government employees or not and the duration of training offered in readiness for the implementation of the interventions. We also categorized the CHW interventions depending on the content and place where interventions were offered. The content of the CHW interventions was focused on either behavioral communication for lifestyle change or the role of CHW as a mediator between patient and health care system. The interventions were either home/community based or clinic based. We also categorized studies based on whether they had reported significant reduction in BP or improvement in the number of participants with controlled BP. The patterns for different CHW interventions and outcomes across different studies was described in the narrative.

Ethics and dissemination

This paper involved a synthesis of publicly available articles and therefore did not require ethical approval.

Patient and Public Involvement

We did not involve patients or members of the public in the design, or conduct, or reporting, or dissemination plans of this research.

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RESULTS

The search strategy identified 1557 records through database searching (n = 1525) and reference lists (n = 32). After excluding duplicates, a total of 1383 records were reviewed (Figure 1). Out of these, 1346 were excluded because information provided in the title and abstract did not meet the inclusion criteria leaving 36 potentially eligible studies for full text review. Of these, 22 did not meet all the inclusion criteria and were excluded for reasons (some more than one), that included not being interventional studies[27-32], not based in low- and middle-income countries[33-42], being protocols of studies [43, 44], wrong study population[45-47] and not reporting on BP as an outcome[27, 34, 48]. Thus, a final 14 studies were used for final data extraction, twelve being cluster randomized trials[49-60] and the other two being interventional pretest and posttest studies[61, 62]. The summary characteristics of the included papers is included in Table 1. The study populations were hypertensive patients in 6 studies[52, 53, 55-57, 60], had at least 1 cardiovascular disease (CVD) risk factor in 2 studies[51, 59] and in 6 studies the participants involved the general population[49, 50, 54, 58, 61, 62]. All the studies involved adult populations except one study that involved children and young people aged 3-39 years[54]. The duration of the interventions ranged from 3 months to 2 years. Geographically, 9 studies were from South Asia[51, 53, 54, 56-61], 3 from Sub Saharan Africa[49, 50, 55] and 1 from East Asia[62] and 1 from South America[52].

Risk of bias for the included studies

The risk of bias of the included studies is shown in Figure 2. The risk of selection bias as result of sequence generation and allocation concealment was low in 12 (86%) of the studies and high in 2(14%). The risk of performance bias and reporting bias due to lack of blinding of participants, personnel and outcome assessors was low in 8(57%) and high in one, but unclear in 5(36%) of the studies. The risk of attrition bias (loss to follow-up) was low in 11(79%), high in one and unclear in two studies while risk of bias due to selective outcome reporting was low in 11(79%) of the studies and unclear in 3(21%).

Table 1. Summary of papers included

Author, year, journal & setting	Study design	Baseline sample characteristics	Intervention design	Outcome measurement	Key findings
Joshi et al, 2019 American Heart Journal (v.216) Rural India	A cluster randomized trial	**controls** 1611 (567 hypertensive), mean age 61.7±10.4 years, mean SBP 130.3±20.8, mean DBP 79.4±11.7, interheart risk score 10; **Intervention** 1650 (597 hypertensive), mean age 61.7±10.2yrs, mean SBP 130.3±20.5, DBP 78.9±11.3, interheart risk score (10)	CHWs home visitation to encourage prescriptions' adherence and measure BP. CHWs provided short goal-directed printed slogans to promote integration of preventive therapies with activities of daily living.	Primary outcome- Mean SBP. Secondary outcome- inter-heart risk score, and adherence to prescription drugs intake	SBP drop was not impacted Inter-heart risk score dropped to 8.2 (intervention) versus 8.3 (control) CHWs promoted adherence to therapy

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Jafar et al. 2009 Annals of Internal Medicine (v.151) Karachi, Pakistan	A cluster randomized trial	1341 hypertensive (SBP ≥140 mmHg or DBP ≥90 mmHg or on antihypertensive and all aged ≥40 years)	Home health education (HHE) by CHWs: 3-monthly HHE, and annual training of general practitioners (GP) in hypertension management.	Primary outcome - change in SBP from baseline to the last follow-up visit (2 years). Secondary outcome - % with <140 mmHg SBP & <90mmHg DBP at last follow up (2 years)	SBP decrease in HHE and GP group (10.8 mm Hg [95% CI, 8.9 to 12.8 mm Hg]) than in the GP-only, HHE-only, or no intervention groups (P<0.001).
Jafar et al, 2020 The New England Journal of Medicine Multicountry: Rural Bangladesh, Pakistan and Sri Lanka	A cluster-randomize controlled trial	2645 hypertensive adults- 1330 intervention group, 1315 control Mean age 58.8±11.5 years 64.3% were female (877 in intervention and 824 in control), 35.7% male (distribution not described), BP was uncontrolled in 69.6% and very poorly controlled in 29.6% of participants SBP (intervention group 146.7±22.4, control group 144.7±21.0 mmHg); DBP (89.1±14.7intervention group and 87.8±13.8 in control)	Home-planned CHWs-led health education (intervention) together with 3-monthly BP monitoring for 24 Months,	Primary outcome - Reduction in SBP at 24 months. Secondary outcomes – Reduction in DBP at 24 weeks. and, % of participants with BP control (SBP<140 mmHg and DBP <90 mmHg)	Mean SBP reduction greater with intervention (by 5.2 mmHg) compared to controls DBP reduction of 2.8 mmHg greater in intervention than in control group. BP control (<140/90 mmHg) achieved in 53.2% of intervention group compared with 43.7% in control group Summary: Linking government trained CHWs with existing public health care infrastructure enhanced BP reduction than in usual care.
Vendatha n et al, 2019 Journal of American College of Cardiology (vol 74) Turbo, Western Kenya	A cluster randomized trial	1460 Participants (58% female, 42% male). **Control** (usual care) 491, **Intervention** paper-based (500), and smartphone (469). Pooled baseline SBP 159.4 mmHg, DBP 89.7±12 mmHg. Usual care (158.4±19.6 & 89.0±12.3), Paper based (158.3±20.3 & 90.4±12.1) and Smart Phone (161.3±18.5 & 89.7±11.5) for SBP & DBP respectively.	Linkage to care intervention through tailored behavioral communication by CHWs using "paper-based" and "smartphone based" approaches.	Primary outcomes - 1) linkage to care, and 2) change in SBP. Secondary outcome – DBP change.	Overall improvement (49%) linkage to care. Significantly greater linkage in the usual care and smartphone arms. Average overall follow-up SBP and DBP were 149.9 and 90.8 mmHg, respectively. Modest improvement in SBP in smart phone than usual care groups (−13.1 mmHg vs. −9.7), **but** difference insignificant. BP control (<140/90) achieved in 26% of participants – Group differences insignificant.
Peiris et al, 2019 PLoS ONE West Godavari District, Andhra Pradesh, Rural India	A stepped-wedge, cluster randomized controlled trial	**Intervention group** (n=4348, 55.6 % Female). 60.3±10.7 years, 46.8% no formal schooling, 45.7% (1986) hypertensive, Angina 11.8%, stroke 9.2%; 10 year CVD risk SBP > 160 mmHg or DBP > 100mmHg 53.4% (2324); on antihypertensives 41.8%; SBP 156.6±22.9; DBP 89.1±13.8 **Control group** (n=4294) 55.9 % Female). 61.0±10.9 years, 46.9% no formal schooling, 46.5% hypertensive (i.e. 1997), Angina 11.8%, stroke 8.1%; 10 year CVD risk SBP >160 mmHg or DBP >100mmHg 52.4% (2251); on BP drugs 42.3%; SBP 157.3±23.2; DBP 88.8±13.9	Three, 6-months intervention thus: (1) mobile tablet based CVD risk assessment by CHWs; (2) electronic referral to PHC doctors; and (3) a tracking system for follow-up care.	Primary outcome – proportion whose SBP drops <140 mmHg. Secondary outcomes – difference in mean BP levels, difference in proportion using BP drugs.	Increased treatment rates among high risk individuals assessed by CHWs, but, the effects on BP were not demonstrated.
Jafar et al, 2010 British Medical Journal	A cluster randomized controlled trial	4023 participants, aged 3-39 years – mean age for HHE, the intervention n=2008 was 18.9±9.8, for control n=2015 was 19.0±9.8; males in HHE 46.3% (i.e 929); Males in	3-monthly home based family health education by CHWs	Primary outcome – change in SBP after 2 years follow-up.	Family based HHE by CHWs reduced the usual rise in BP with age in children and young adults.

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Karachi, Pakistan		control 45.8% (i.e 923); mean SBP in HHE 114±14 mmHg, in control 115±14; mean DBP in HHE 74±11, in control also 74±11. HHE literacy 11.5%, in non HHE 13.8%		Secondary outcome – Change in DBP between baseline and at 2 years	
He et al, 2017 The Journal of the American Medical Association Argentina (area not specified)	A cluster randomized trial	1432 hypertensive adults Mean age intervention group (n=743) age 56.1±13.6 years, control (n=689) 55.5±13.0 years. 52.6% and 53.4% of intervention and control groups respectively were female, Mean SBP 151.7±16.8 and 149.8±15.5 mmHg and DBP, 92.2±12.2 and 90.1±12.9 among intervention and control groups respectively. 86% in intervention group and 84% in control group were on BP drugs	multicomponent intervention: (1) CHW–led home health coaching, BP monitoring, BP audit and feedback, and (2) a GP intervention, and a (3) text messaging intervention over 18 months. Controls receiving usual care.	Primary outcomes of differences in SBP and DBP between baseline and follow-up end Secondary outcomes were % with controlled BP <140/90mmHg	CHWs helped low-income patients with uncontrolled hypertension achieve greater decrease in SBP and DBP through multicomponent interventions compared to patients receiving usual care over 18 months.
Khetan etal 2019 Global Heart journal Dalkhola, India	A Cluster Randomized Controlled Trial	1242 adults (35 to 70 years drawn from 12 clusters with at least 1 CVD risk factor (hypertension 650, diabetes 317, smoking 500) Intervention group- 736 Mean age 52.1±9.6 years Control group- 506 Mean age- 51.7 ± 9.8 years.	Behavioral change communication and BP monitoring through regular CHWs home visits once every 2 months for 2 years. The control group received usual care in the community	Primary outcomes were change in SBP, FBG and self-reported number of daily cigarettes from visit 1 to post-intervention among hypertensives, diabetics and smokers.	Improved SBP in hypertension, and inconclusive effect on FBG in diabetes, and no demonstrable effect on smoking.
Setiawan, D & Sargowo ,D 2018 European Society of cardiology congress Batu City, Indonesia	Interventional pre and post-test study	102 subjects (Men 21; Women 81) aged 30 to 75. Mean Framingham Risk Score (FRS) of 6.17 65.75	Community based primary intervention that included 60-minutes of heart exercise thrice weekly for 3-months, and regular health counselling and education on lifestyle changes	Primary outcome – change in mean value of FRS	A community-based health education and physical activity program significantly improved the FRS of a farmer's population. Mean FRS changed from 6.17 65.75 to 4.16 63.92 post-intervention group (Z score = -6.009; p =0.00).
Sankaran et al 2016 Journal of general internal medicine Rural region of Southern India	Interventional pre and post study	598 patients over the age 60 were screened. Baseline hypertension prevalence rate of 44%.	CHW trained for 7 months offered a community intervention for 1 year, that involved diagnosis, management and referral of patients with uncontrolled BP and provision of continuing medications and lifestyle advice to those with well-controlled hypertension. GPs did initial evaluation and medication while nurses supervised CHWs, weekly.	Primary outcome- BP control at 1 year	After 1 year, 51.8% of hypertensive patients had BP <140/90 mmHg. A teaching program educating community health workers about hypertension followed by screening, diagnosis, and management of hypertension by CHW in the field is a feasible mechanism for achieving hypertension control in a remote tribal community in India.
Gamage et al 2020 Plos One-Medicine	A cluster randomized controlled trial	2,382 hypertensive patients. Intervention group- 637 from 5 clusters Mean age-56.6±14.3 Mean SBP 140.5±22.7 Mean DBP 80.4±13.7 Controlled hypertension- 43.5	Trained CHWs delivered group-based intervention to hypertensive people, consisting of 6 fortnightly sessions of ~90	Primary outcome-change from baseline in the proportion of people with controlled hypertension (BP < 140/90 mm Hg)	BP declined an average of 5.0/2.1 mm Hg more in the intervention group than the usual care group, and control of BP improved from baseline to follow-up, more in the intervention (from 227 [49.5%] to 320 [69.7%] individuals) than

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South India		On hypertensive drugs-38% Control group-1097 form 10 clusters Mean age-56.9 Mean SBP 137.8 ±22.2 Mean DBP 80.6 ±13.9 Controlled hypertension-50.1% On hypertensive drugs -40.6%	minutes held within participants' villages (clusters). Sessions included BP monitoring, education about hypertension, and support for healthy lifestyle change.	Secondary outcome- change in SBP and DBP from baseline	in the control group (from 528 [52.2%] to 624 [61.7%] individuals) (odds ratio [OR] 1.6, 95% CI 1.2–2.1; P = 0.001)
Neupane et al 2018 Lancet Glob Health Nepal	An open-label, cluster-randomized trial	1468 with mean BP of 122/80 mm Hg. 16% of participants were smokers, 11% were drinking alcohol in amounts harmful to their health, and 5% had low physical activity. Intervention group 425 normotensive 175 pre-hypertensive 255 hypertensive Control 305 normotensive 128 pre-hypertensive 180 hypertensive	A lifestyle intervention led by female community health volunteers (FCHVs) or usual care (control group). 43 FCHVs provided home visits every 4 months for lifestyle counselling and BP monitoring for a period of 1 year.	Primary outcome- mean systolic blood pressure at 1 year.	The intervention was effective for reduction of BP in individuals with hypertension and ameliorates age-related increases in BP in adults without hypertension in the general population. The mean SBP at 1 year was significantly lower in the intervention group than in the control for all cohorts (–2·28 mm Hg (95% CI –3·77 to –0·79, p=0·003) for normotensive participants, –3·08 mm Hg (–5·58 to –0·59, p=0·015) for prehypertensive participants, and –4·90 mm Hg (–7·78 to –2·00, p=0·001) among hypertensives.
Cappuccio et al 2006 BMC Public Health Ashanti region of central Ghana	A community-based cluster randomized trial	1,013 participants from 12 villages (628 women, 481 rural dwellers). Intervention group Mean age-54±11 years Mean BMI- 21±4 Mean SBP- 129±25 Mean DBP- 77± 13 Hypertensive 30% Control group Mean age-55±11 years Mean BMI- 21±4 Mean SBP- 127±27 Mean DBP- 76± 13 Hypertensive –28 %	Community-based trial of health promotion for population-wide reduction in salt intake, to lower population BP. Health promotion targeted villagers willing to attend and was provided through daily sessions in week one followed by 6-months' once weekly sessions in communal places (church & schools)	Primary outcome- change in mean BP and urinary sodium.	At six months the intervention group showed a reduction in systolic (2.54 mmHg [-1.45 to 6.54]) and diastolic (3.95 mmHg [0.78 to 7.11], p = 0.015) BP when compared to control. There was no significant change in UNa.
Goudge et al 2017 BMJ Global health Mpumalanga Province, South Africa	A pragmatic cluster randomized controlled trial with repeated cross-sectional surveys.	Control group - 1908 adults with mean age of 56.4±19.8 years, females -56% No hypertension-53.7% Controlled hypertension on medication- 10.2% Uncontrolled hypertension on medication- 9.2% Not on treatment- 27% Intervention group - 1505 adults Mean age-56.81±9.7 years Females -55% No hypertension-53% Controlled hypertension on medication- 6.6% Uncontrolled hypertension on medication- 8.8% Not on treatment- 31.6%	Intervention consisted of two LHWs assigned to each clinic to support nurses over for 18 months in management of patients with chronic disease through booking appointments, retrieving and filing patient files, health education, taking vital signs, prepacking of medication and texting reminders for appointments	Primary outcome - change of proportion of the population who had uncontrolled hypertension with a risk profile indicating at least moderate risk of CVD. Secondary outcomes - changes in proportion of the population with undiagnosed hypertension, had their BP measured, were using medication for hypertension and retention in care of people with diagnosed hypertension defined.	There was no reduction in those with moderate or greater CVD risk but there was improved adherence. The intervention did not improve BP control, despite its success in increasing the number of patients with hypertension attending the clinic, as well as the number that attended on their appointed day.

CVD-Cardiovascular disease; CHWs- Community health workers; BP- Blood pressure; BMI- Body mass index DBP- Diastolic blood pressure; SBP- Systolic Blood pressure; FBG- Fasting blood glucose; FCHVs- Family community health volunteers; Framingham Risk Score; GP- General practitioner; HHE-Home health education; LHWs- Lay health workers; UNa –Urinary sodium

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Recruitment and training of CHW

Given the nature of CHW intervention for management of hypertension, the CHW acted as both provider and recipients of the intervention. We therefore examined the recruitment and training requirements for CHW applied in different studies. While all the studies met the inclusion criteria of CHW defined as a lay health worker with basic training to supply community members access to health and social services, the recruitment and training of the same was different across the studies. In all of the studies CHW were persons with at least 8 to 10 years of education drawn from the local community but in some of the studies [58, 61] the CHW were specifically female. Most of the studies [55-57, 59] utilized government CHW already working in the health care system while other studies [50, 51, 53, 54, 58, 60] recruited CHW specifically for the study following same criterion used in recruiting the Government CHWs. There was no standard training for the CHW across the different studies in this review. The CHW underwent various forms of training to be able to carry out their roles in the studies. The training period ranged from 2 days - 6 weeks with ongoing support from the health care workers. In some of the studies the trainings were shorter ranging from 2-5 days [52, 56, 58, 59] while others ranged from 4-6 weeks [51, 53, 54, 60, 61]. In some of the studies the training period was staggered throughout the study period [51, 57, 61]. In one of the studies the length of the training period for CHW was not specified [55].

Similarly, the content of the training was different across the studies. The content of training included, home health education on CVD risks and behavior change communication strategies [55, 57, 58, 60], basics of hypertension and assessments of CVD risk factors to include measurements technique for BP monitoring and anthropometric measurements [51-53, 56, 58, 59, 61], as well as referral procedure for hypertensive patients [58] and survey methods [53]. The recruitment and the CHW training details was not reported in 2 of the studies [49, 50].

Content and Approaches of the CHWs interventions.

The CHWs intervention were designed to promote BP control mainly for hypertensive patients

and those with CVD risks but also targeted prevention of high BP in healthy individuals. The interventions were either home-based or community-based with an exception of one study [50] which incorporated the use of CHWs in the primary health clinic to assist nurses in the management of hypertension.

The content of the activities implemented by the CHWs during the interventions was similar across many of the studies. In addition to the screening or monitoring of hypertension through BP measurements and audits which was the main outcome measure in this review, the CHW were also involved in other activities geared to the management and control of hypertension. First, the major activity in the CHW interventions as outlined in table 1 was health promotion through home-based health education and lifestyle counselling[51-55, 57, 58, 60] and community level behavior change communication[49, 56, 61, 62]. Health education included teaching about hypertension, motivation to engage in care and support for healthy lifestyle change such as reduction in salt intake. Secondly, in one of the studies [59] CHW were involved in CVD risk assessment using a mobile tablet-based CVD risk assessment tool. Thirdly, in addition to the health education and BP monitoring, one trial incorporated physical activity component of 60 minutes of heart exercise

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three times a week for a period of 3 months[62]. Fourth, the CHWs provided follow-up for hypertensive patients and supported patients by monitoring treatment adherence in one of the trials[53]. Finally, in some studies CHWs were utilized as mediators between patients and the healthcare system by providing referral and encouraging linkage to care for those requiring hypertensive treatment [55, 59, 61].

Outcomes and main findings of the studies

The outcomes for the CHWs interventions were different for various studies as shown in Table 1. The reported primary outcome in most of the studies[49, 51-55, 57, 58] was changes in mean SBP while other studies [50, 56, 59, 61] reported proportion of participants with controlled BP (SBP<140 mmHg and DBP <90 mmHg) as a primary outcome. There was significant reduction in the mean SBP in seven studies[49, 51, 52, 54, 57, 58, 60] while in two of the studies [53, 55] the reduction in mean SBP was insignificant. Two of the RCTs reported a reduction in the usual rise in BP with age among children and young adults[54] and normotensive adults in the general population[58]. Similarly, three studies [56, 57, 61] reported significant improvement in the proportion of participants with controlled BP as result of the CHW interventions while three other studies[50, 55, 59] did not find any significant improvement in the BP control for participants. It is notable that even for studies which reported negative results in the control of BP, there were positive effect on linkage to care[55, 59], treatment adherence [50, 53] and inter-heart risk score[53]. In other studies there was improvement in the primary outcome of mean change in Framingham risk score(FRS) [62], and reduction in the proportion of patients with moderate or high CVD risk level[50].

Other outcomes reported in the studies were changes in DBP[55-57, 60], changes in urinary sodium[49], proportion of the population with undiagnosed hypertension, proportions who had their BP measured and those retained in care among the diagnosed hypertensive patients [50], as well as difference in the proportion of patients on hypertension treatment [50, 59]. In one RCT that had participants with at least one CVD risk factor, FBS among diabetes patients and mean number of daily cigarettes among smokers were reported as primary outcomes in addition to change in SBP[51]. In this study, CHWs interventions had a positive effect on SBP but had inconclusive effect on FBS and mean number of cigarettes smoked[51].

Outcomes trends in relation to the implementation strategy of the CHW intervention.

Studies whose CHW intervention focused on community/home based behavioral change communication[49, 51, 52, 54, 57, 58, 60] had significant reduction in SBP and significant improvement in the proportion of participants with controlled BP[56, 57, 61]. On the other hand, interventions that focused on the role of CHW as mediators between patient and health care system through referral, linkage to care and monitoring drug adherence[50, 53, 55, 59] did not have significant impact on BP control.

Although the studies lacked a standard requirement for CHW training in terms of duration and content, training was an important component of the CHW intervention for management and control of hypertension. Studies that had training ranging from 4-6 weeks had positive outcomes

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with those that staggered the trainings across the entire intervention period all showing positive outcomes[51, 57, 61] in the control and management of high BP.

DISCUSSION

Weak health care systems and a rising burden of CVD calls for health system innovations in the management of CVD in LMICs. CHWs have been suggested as a way to achieve that goal, with their potential ability to quickly integrate into the health care system. With short training periods, CHW can provide community-based care that is cost-effective and achieve high-quality outcomes [7]. Our review examined the nature and the role CHWs can play in management of hypertension in LMIC. This review found that, in general, studies that linked CHWs with existing public health care infrastructure enhanced control and management of BP over and above usual care. The CHW interventions had positive effects in the reduction of BP, linkage to care, treatment adherence, and in reducing CVD risk level among both hypertensive and normotensive individuals in LMICs. This underscores the important role CHWs would play in the management of hypertension yet at a cheaper cost in countries with leaner economic muscle[63-65]. The review further showed that CHW play diverse roles ranging from preventive, therapeutic and health systems utilization in management of hypertension.

With the growing burden of NCDs, primary prevention is a major pillars in the control of these diseases in LMICs[66]. In light of critical shortages in the health workforce in LMICs, CHWs provide cheaper alternative that would form the backbone of most primary health care(PHC) services for management and control of both communicable and NCDs[67]. The studies in this review demonstrate that task-shifting of roles such as health education, interpersonal communication on lifestyle modifications with a focus on primary prevention, screening for early diagnosis and supporting self-management behavior can lead to significant reduction of BP. Studies from developing countries show that CHWs have high acceptability at local community level which, may contribute to the dual benefit of providing both cheaper and acceptable labor in addition to the direct contribution in BP control [68].

While trained CHW may not perform in the same capacity as trained nurses and health educators, with appropriate and specifically focused training and supervision they can successfully contribute to the management of hypertension, as demonstrated by many of the studies reviewed. In the current review, we did not find any standards in the training of CHW for management of hypertension. The training duration ranged from 2 days to 6 weeks and the content was different across the studies depending on the focus of the intervention. The reviews showed that CHWs intervention entailed screening, promotion and monitoring of BP and its control strategies at both home and community levels where only basic PHC approaches were required. The basics required here could be easy and faster to teach with no need for unnecessary scientific details and the favorable study results from the current review show that such training as the CHWs received is enough to yield demonstrable behavior change and BP modulation outcomes[49-62].

There is general agreement in the reviewed literature that CHWs can contribute to significant reduction of BP and hypertension as a key modifiable risk factor for CVD and premature death [1, 2], especially in the LMICs[3]. This review shows that enjoining CHWs in the screening and care of hypertensive patients from such countries has positive results in care of such patients. Most of

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the studies in this review had low risk of selection bias in both sequence generation and allocation concealment through randomization (11 studies), had low performance bias because they had both participants and investigators blinded (8 studies), had blinding of assessors and so detection bias was minimal (9 studies), and had selective reporting bias minimized at conceptualization and implementation of the protocol as opposed to allowing a direct influence of the outcomes at dissemination stage (11 studies) strengthens the evidence of our review. As such, the evidence deduced on importance of CHWs in mitigation of hypertension becomes especially critical for countries with less resources, an ageing population, an increase in sedentary lifestyles and other lifestyle risk factors, and a concomitant direct increase in proportions of the general population who are hypertensive[2]. The literature suggests a way out of the increasing incidence of hypertension in these countries and rising proportion of persons with poorly controlled BP. The CHWs drawn from these communities would advance the care needed and supplement the lean professional workforce available. CHWs have cultural understanding which is crucial in providing the necessary linkage between community and health care system [13]. The current review also demonstrates that their roles can be diverse yet effective in the control and management of high BP and other cardiovascular risk factors.

Study limitations and strengths

Our study represents a scope review of approaches and outcomes specific to CHW interventions in the management and control of hypertension in LMICs. The strengths of the current study are that we conducted comprehensive searches of databases to ensure that all relevant publications were identified. We also reduced potential bias in the review by having the authors independently screen through the search results and extract the data autonomously. However, the study is limited in that, due to diversity of the study participants, interventions, and outcomes of the studies included in the review, a quantitative synthesis (meta-analysis) of the overall effect of CHW interventions was not possible and therefore outcomes were summarized for individual studies.

Implications of the results for practice, policy and future research

This study showed that CHWs drawn from community are an important resource in the management and control of hypertension. Preference should be given to CHW interventions focused on behavior change communication and lifestyle counselling for reduction of high BP for both hypertensive and normotensive individuals. There is a need for adoption of standard curricula for training of CHW for the control and management of hypertension to guide translation of such interventions in different settings in LMICs. Future reviews should look into the overall effect of the various components of CHW interventions and their cost-effectiveness in the management and control of CVDs.

Conclusions

In conclusion the review suggests that CHWs interventions linked with support and/or supervision from health care workers provides a promising avenue for achieving improvements in hypertension control in LMICs. There is need to adopt this and integrate CHW community-based lifestyle interventions in PHC for overall reduction of CVD risks.

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OTHER INFORMATION

Protocol and registration: Methods of the review and inclusion were documented in advance, but not published in a review protocol.

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Conflicts of Interest

The authors declare that they have no competing interests.

Availability of data and materials

No additional data are available.

Authors contribution

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GWM conceived the study idea. GWM and KM performed the literature search. GWM and KM performed data extraction and evidence synthesis. GWM wrote the first draft of the manuscript. JAP and KM critically revised the manuscript. GWM, KM and JAP approved the final version of the manuscript.

Figure caption

Figure 1:PRISMA flow chart

Figure 2:Risk for bias assessment

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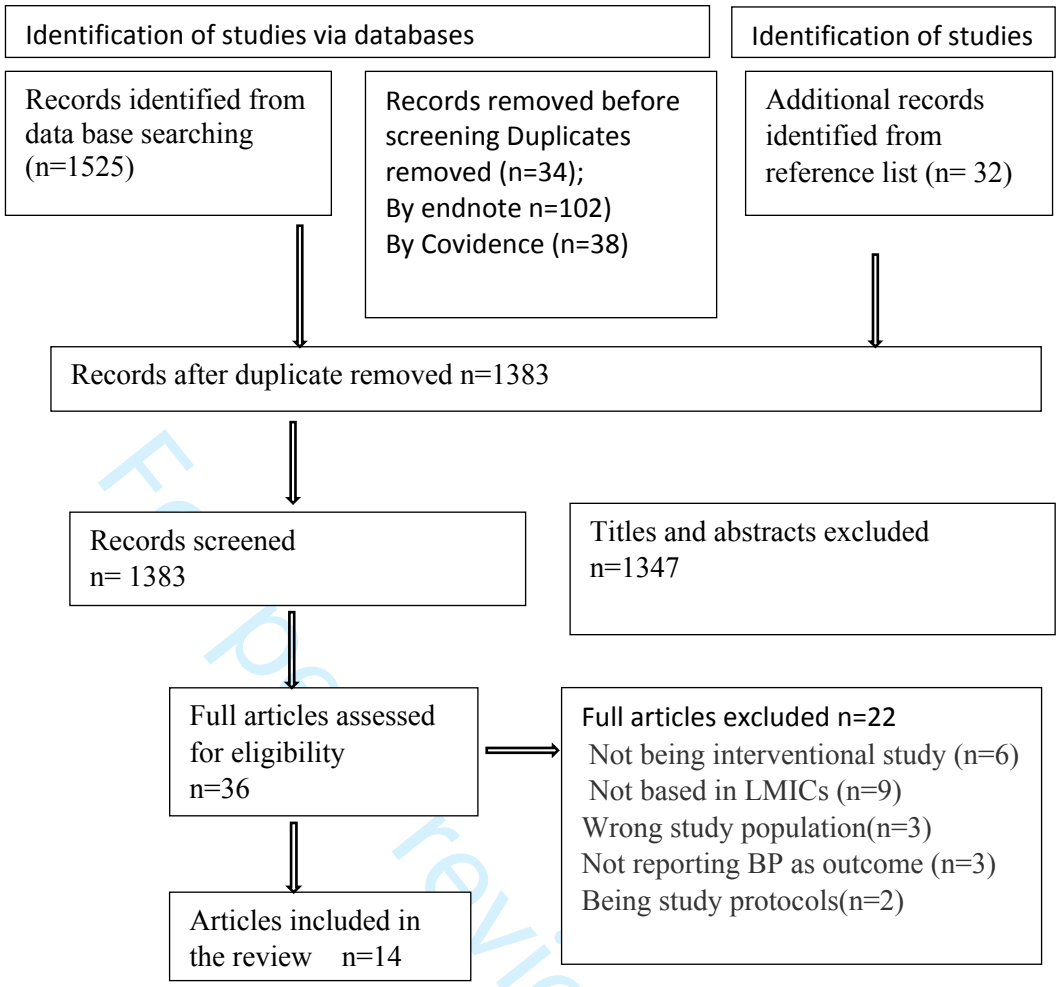


Figure 1. PRISMA Flow diagram showing study selection

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	Sequence generation (selection bias)	Allocation concealment (selection bias)	Blinding of participants and personnel for all outcomes(performance bias)	Blinding outcome assessors(detection bias)	Incomplete outcome data for all outcomes (attrition bias)	Selective outcome reporting(reporting bias)
Cappuccio et al 2006	-	+	+	+	-	+
Gamage et al 2020	+	+	+	-	+	+
Goudge et al 2018	+	+	?	?	+	+
He et al 2017	+	+	+	+	+	+
Joshi et al 2019	+	+	+	+	+	+
Jafar 2009	+	+	+	+	-	?
Jafar 2010	+	?	+	+	-	+
Jafar et al 2020	+	+	+	+	+	+
Khetan et al 2019	+	+	?	+	+	+
Neupane 2018	+	+	-	?	+	?
Peiris et al 2019	+	+	+	+	+	+
Setiawan, D & Sargowo, D 2018	-	-	?	+	?	?
Sankran et al 2016	-	-	?	?	?	+
Vedanthan et al 2019	+	+	?	?	+	+

- High risk of bias
+ Low risk of bias
? Unclear risk of bias

Figure 2: Risk for bias assessment

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Reporting checklist for systematic review (with or without a meta-analysis).

Based on the PRISMA guidelines.

Instructions to authors

Complete this checklist by entering the page numbers from your manuscript where readers will find each of the items listed below.

Your article may not currently address all the items on the checklist. Please modify your text to include the missing information. If you are certain that an item does not apply, please write "n/a" and provide a short explanation.

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In your methods section, say that you used the PRISMAreporting guidelines, and cite them as:

Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, Shamseer L, Tetzlaff JM, Akl EA, Brennan SE, Chou R, Glanville J, Grimshaw JM, Hróbjartsson A, Lalu MM, Li T, Loder EW, Mayo-Wilson E, McDonald S, McGuinness LA, Stewart LA, Thomas J, Tricco AC, Welch VA, Whiting P, Moher D. The PRISMA 2020 statement: An updated guideline for reporting systematic reviews

		Reporting Item	Page Number
Title			
Title	#1	Identify the report as a systematic review	1
Abstract			

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Abstract	#2	Report an abstract addressing each item in the PRISMA 2020 for Abstracts checklist	1-2
Introduction			
Background/rationale	#3	Describe the rationale for the review in the context of existing knowledge	3
Objectives	#4	Provide an explicit statement of the objective(s) or question(s) the review addresses	4
Methods			
Eligibility criteria	#5	Specify the inclusion and exclusion criteria for the review and how studies were grouped for the syntheses	4
Information sources	#6	Specify all databases, registers, websites, organisations, reference lists, and other sources searched or consulted to identify studies. Specify the date when each source was last searched or consulted	5
Search strategy	#7	Present the full search strategies for all databases, registers, and websites, including any filters and limits used	5
Selection process	#8	Specify the methods used to decide whether a study met the inclusion criteria of the review, including how many reviewers screened each record and each report retrieved, whether they worked independently, and, if applicable, details of automation tools used in the process	5

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Data collection process	#9	Specify the methods used to collect data from reports, including how many reviewers collected data from each report, whether they worked independently, any processes for obtaining or confirming data from study investigators, and, if applicable, details of automation tools used in the process	5
Data items	#10a	List and define all outcomes for which data were sought. Specify whether all results that were compatible with each outcome domain in each study were sought (for example, for all measures, time points, analyses), and, if not, the methods used to decide which results to collect	5
Study risk of bias assessment	#11	Specify the methods used to assess risk of bias in the included studies, including details of the tool(s) used, how many reviewers assessed each study and whether they worked independently, and, if applicable, details of automation tools used in the process	5
Effect measures	#12	Specify for each outcome the effect measure(s) (such as risk ratio, mean difference) used in the synthesis or presentation of results	N/A
Synthesis methods	#13a	Describe the processes used to decide which studies were eligible for each synthesis (such as tabulating the study intervention characteristics and comparing against the planned groups for each synthesis (item #5))	5-6

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Synthesis methods	#13b	Describe any methods required to prepare the data for presentation or synthesis, such as handling of missing summary statistics or data conversions	N/A
Synthesis methods	#13c	Describe any methods used to tabulate or visually display results of individual studies and syntheses	5-6
Synthesis methods	#13d	Describe any methods used to synthesise results and provide a rationale for the choice(s). If meta-analysis was performed, describe the model(s), method(s) to identify the presence and extent of statistical heterogeneity, and software package(s) used	5
Synthesis methods	#13e	Describe any methods used to explore possible causes of heterogeneity among study results (such as subgroup analysis, meta-regression)	N/A
Synthesis methods	#13f	Describe any sensitivity analyses conducted to assess robustness of the synthesised results	N/A
Reporting bias assessment	#14	Describe any methods used to assess risk of bias due to missing results in a synthesis (arising from reporting biases)	N/A
Certainty assessment	#15	Describe any methods used to assess certainty (or confidence) in the body of evidence for an outcome	N/A
Data items	#10b	List and define all other variables for which data were sought (such as participant and intervention characteristics, funding sources). Describe any	5

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		assumptions made about any missing or unclear information	
Results			
Study selection	#16a	Describe the results of the search and selection process, from the number of records identified in the search to the number of studies included in the review, ideally using a flow diagram (http://www.prisma-statement.org/PRISMAStatement/FlowDiagram)	6
Study selection	#16b	Cite studies that might appear to meet the inclusion criteria, but which were excluded, and explain why they were excluded	6
Study characteristics	#17	Cite each included study and present its characteristics	6
Risk of bias in studies	#18	Present assessments of risk of bias for each included study	6
Results of individual studies	#19	For all outcomes, present for each study (a) summary statistics for each group (where appropriate) and (b) an effect estimate and its precision (such as confidence/credible interval), ideally using structured tables or plots	7-10
Results of syntheses	#20a	For each synthesis, briefly summarise the characteristics and risk of bias among contributing studies	N/A
Results of syntheses	#20b	Present results of all statistical syntheses conducted. If meta-analysis was done, present for each the summary	N/A

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		estimate and its precision (such as confidence/credible interval) and measures of statistical heterogeneity. If comparing groups, describe the direction of the effect	
Results of syntheses	#20c	Present results of all investigations of possible causes of heterogeneity among study results	N/A
Results of syntheses	#20d	Present results of all sensitivity analyses conducted to assess the robustness of the synthesised results	N/A
Risk of reporting biases in syntheses	#21	Present assessments of risk of bias due to missing results (arising from reporting biases) for each synthesis assessed	N/A
Certainty of evidence	#22	Present assessments of certainty (or confidence) in the body of evidence for each outcome assessed	N/A
Discussion			
Results in context	#23a	Provide a general interpretation of the results in the context of other evidence	12
Limitations of included studies	#23b	Discuss any limitations of the evidence included in the review	13
Limitations of the review methods	#23c	Discuss any limitations of the review processes used	13
Implications	#23d	Discuss implications of the results for practice, policy, and future research	14
Other information			

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Registration and protocol	#24a	Provide registration information for the review, including register name and registration number, or state that the review was not registered	20
Registration and protocol	#24b	Indicate where the review protocol can be accessed, or state that a protocol was not prepared	20
Registration and protocol	#24c	Describe and explain any amendments to information provided at registration or in the protocol	N/A
Support	#25	Describe sources of financial or non-financial support for the review, and the role of the funders or sponsors in the review	20
Competing interests	#26	Declare any competing interests of review authors	20
Availability of data, code, and other materials	#27	Report which of the following are publicly available and where they can be found: template data collection forms; data extracted from included studies; data used for all analyses; analytic code; any other materials used in the review	20

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